

- 2 (a) Diabetes insipidus (DI) is a condition in humans that causes a person to have an excessive thirst, which leads to increased drinking. One form of DI is caused by a tumour in the region of the hypothalamus concerned with osmoregulation.

(i) Explain how a tumour develops in the hypothalamus.

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..... [2]

(ii) Suggest how a tumour in the hypothalamus can lead to a person producing a large volume of dilute urine.

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..... [3]

- (b) The nervous and endocrine systems of a mammal co-ordinate responses to changes in the internal and external environment. Table 2.1 compares some features of the nervous and endocrine systems.

Complete Table 2.1.

Table 2.1

feature	nervous system	endocrine system
signal	impulse	
method of transmission of signal		in blood
type of communication		chemical
duration of effect		

[4]

[Total: 9]